

FGFR3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR3 gene fusions from the msk_impact_50k_2026 study, covering 152 fusion events. 74/152 fusions (48.7%) were in-frame. 142/152 fusions (93.4%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0194824$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 152 analyzed fusion events, identifying 15 distinct fusion partner genes. The most frequent partner was TACC3, observed in 131 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 73/74 (98.6%) of in-frame fusions retained the domain, $p=0.0194824$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Domain-disruption analysis was skipped: FGFR3 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 151 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 677 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -71 aa from the nearest configured domain boundary at 748 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: FGFR3 has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: FGFR3 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TACC3	131	0.8618421052631579	0.20043213737826426	None	0.8507823563952885	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.6232543134040097	1
TNIP2	3	0.019736842105263157	0.20043213737826426	None	0.8763197586726998	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26782435825292855	2
CCDC149	2	0.013157894736842105	0.20043213737826426	None	0.8778280542986425	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2653280155863073	3
JAKMIP1	2	0.013157894736842105	0.20043213737826426	None	0.8778280542986425	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2653280155863073	4
LPCAT2	2	0.013157894736842105	0.20043213737826426	None	0.8770739064856712	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26516641248352774	5
PTBP1	1	0.006578947368421052	0.20043213737826426	None	0.8778280542986425	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2625084667141268	6
CTBP1	1	0.006578947368421052	0.20043213737826426	None	0.8763197586726998	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26218526050856766	7
MAEA	1	0.006578947368421052	0.20043213737826426	None	0.8506787330316742	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2566907550140622	8
ROBO1	1	0.006578947368421052	0.20043213737826426	None	0.8159879336349924	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24925701228620178	9
NSD2	3	0.019736842105263157	0.20043213737826426	None	0.6561085972850679	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22063625224129316	10
RBM20	1	0.006578947368421052	0.20043213737826426	None	0.2413273001508296	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1261154479681669	11
BABAM2	1	0.006578947368421052	0.20043213737826426	None	0.20965309200603321	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11932811765142481	12
SLIT2	1	0.006578947368421052	0.20043213737826426	None	None	None	['domain_retention', 'recurrence']	0.0946940337365316	13
GRK4	1	0.006578947368421052	0.20043213737826426	None	0.0015082956259426794	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07472566128426256	14

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
DGKQ	1	0.006578947368421052	0.20043213737826426	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07440245507870341	15

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
14	1	141	0	9	None	1.0	-0.0
15	2	140	0	9	None	1.0	-0.0
42	2	140	1	8	0.11428571428571428	0.1694208631494733	0.7710331099865381
127	3	139	1	8	0.17266187050359713	0.21992905390389725	0.6577173940286984
153	4	138	1	8	0.2318840579710145	0.26768849958325047	0.5723702865449334
174	4	138	2	7	0.10144927536231885	0.04197605082466328	1.3769984229650098
188	4	138	3	6	0.057971014492753624	0.004616748961172982	2.33566373986112
311	4	138	4	5	0.036231884057971016	0.00037923555536042514	3.421090951996021
376	4	138	5	4	0.02318840579710145	2.3640024802728035e-05	4.626352072135864
648	4	138	6	3	0.014492753623188406	1.1022799082261042e-06	5.957708108603845
653	4	138	7	2	0.008281573498964804	3.6665492410410534e-08	7.4357424781390336
677	4	138	8	1	0.0036231884057971015	7.825580002902196e-10	9.106483464295888
753	5	137	8	1	0.004562043795620438	2.021680915172072e-09	8.694287388663584
758	52	90	8	1	0.07222222222222222	0.002807318578004277	2.5517083002909002
759	88	54	8	1	0.2037037037037037	0.15623087834924132	0.806233125571681
760	89	53	8	1	0.2099056603773585	0.15840511374693833	0.8002308023157874
762	91	51	8	1	0.22303921568627452	0.16500164368454323	0.7825117294856598
763	93	49	8	1	0.2372448979591837	0.27269821801700755	0.5643177000044979
764	95	47	8	1	0.2526595744680851	0.2731204045238703	0.5636458529869772
766	96	46	8	1	0.2608695652173913	0.27473748927465147	0.5610820749288234
767	97	45	8	1	0.26944444444444443	0.27730934214956404	0.5570354994244366
768	100	42	8	1	0.2976190476190476	0.4468103647595553	0.34987676101374454
769	101	41	8	1	0.3079268292682927	0.4455794616522395	0.3510748355557522
770	105	37	8	1	0.3547297297297297	0.45054685984922543	0.34626003284121815
771	108	34	8	1	0.39705882352941174	0.6854647233823904	0.16401489074257258
772	111	31	8	1	0.4475806451612903	0.6849966189342125	0.16431157213069608
773	113	29	8	1	0.4870689655172414	0.688608956187972	0.16202733307235542

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
776	115	27	8	1	0.5324074074074074	1.0	-0.0
777	119	23	8	1	0.6467391304347826	1.0	-0.0
778	120	22	8	1	0.6818181818181818	1.0	-0.0
779	121	21	8	1	0.7202380952380952	1.0	-0.0
780	122	20	8	1	0.7625	1.0	-0.0
781	125	17	8	1	0.9191176470588235	1.0	-0.0
782	127	15	8	1	1.0583333333333333	1.0	-0.0
785	128	14	8	1	1.1428571428571428	1.0	-0.0
786	130	12	8	1	1.3541666666666667	0.5655253555991285	0.24754791850170255
787	131	11	9	0	0.0	1.0	-0.0
788	133	9	9	0	0.0	1.0	-0.0
789	134	8	9	0	0.0	1.0	-0.0
793	138	4	9	0	0.0	1.0	-0.0
797	139	3	9	0	0.0	1.0	-0.0
799	140	2	9	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

73	69
1	8

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	151	142	3	6	0.22663457681508226

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BABAM2	1
CCDC149	2
CTBP1	1

Partner_gene	Event_count
DGKQ	1
GRK4	1
JAKMIP1	2
LPCAT2	2
MAEA	1
NSD2	3
PTBP1	1
RBM20	1
ROBO1	1
SLIT2	1
TACC3	131
TNIP2	3

Per-event results (results.tsv)

event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _sta tus	tar get _ro le	break point_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-000 0025-T03-I M6-1	P-0000 025-T0 3-IM6		DGKQ -FGF R3	DG KQ	unk now n	thr ee _pr im e	179570 1	1	14	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			DGKQ (NM_001347) - FGFR3 (NM_000142) rearrangement: c.352-750: DGKQ_c.40:FGFR3inv Protein Fusion: mid-exon {DGKQ:FGFR3}	NA
EVT-P-002 0773-T01-I M6-2	P-0020 773-T0 1-IM6		FGFR 3-BAB AM2	BAB AM2	unk now n	thr ee _pr im e	180310 5	4	153	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			BRE (NM_004899) - FGFR3 (NM_000142) rearrangement: t(2;4)(p23. 2;p16.3)(chr2:g.28322454: :chr4:g.1803105) Protein Fusion: mid-exon {BRE:FGFR3}	NA
EVT-P-005 4353-T02-I M7-4	P-0054 353-T0 2-IM7		FGFR 3-CC DC14 9	CC DC1 49	in-fr ame	five _pr im e	180870 1	16	758	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - CCDC149 (NM_001130726) fusion: c .2274+40:FGFR3_c.490-9 9:CCDC149inv Protein Fusion: in frame {FGFR3:CCDC149}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 4353-T01-I M6-5	P-0054 353-T0 1-IM6		FGFR 3-CC DC149	CC DC149	in-frame	five_prime	1808701	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - CCDC149 (NM_001130726) fusion: c.2274+40:FGFR3_c.490-99:CCDC149inv Protein Fusion: in frame {FGFR3:CCDC149}	NA
EVT-P-005 7559-T02-I M7-6	P-0057 559-T0 2-IM7		FGFR 3-CTBP1	CTBP1	in-frame	five_prime	1808812	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - CTBP1 (NM_001328) fusion: c.2275-31:FGFR3_c.196-1497:CTBP1inv Protein Fusion: in frame {FGFR3:CTBP1}	NA
EVT-P-001 9022-T02-I M6-35	P-0019 022-T0 2-IM6		FGFR 3-GRK4	GRK4	unknown	three_prime	1795705	1	15	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			GRK4 (NM_182982) - FGFR3 (NM_000142) rearrangement: c.339+3191:GRK4_c.44:FGFR3dup Protein Fusion: mid-exon {GRK4:FGFR3}	NA
EVT-P-000 2610-T01-I M3-36	P-0002 610-T0 1-IM3		FGFR 3-JAKMIP1	JAKMIP1	in-frame	five_prime	1808734	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (FGFR3-JAKMIP1)	NA
EVT-P-003 4419-T01-I M6-37	P-0034 419-T0 1-IM6		FGFR 3-JAKMIP1	JAKMIP1	in-frame	five_prime	1808749	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - JAKMIP1 (NM_001099433) fusion (FGFR3 exons 1-17 fused to JAKMIP1 exons 4-21): c.2274+88:FGFR3_c.624+9161:JAKMIP1inv Protein Fusion: in frame {FGFR3:JAKMIP1}	NA
EVT-P-006 8474-T01-I M7-39	P-0068 474-T0 1-IM7		FGFR 3-MAEA	MAEA	unknown	five_prime	1808895	17	776	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - MAEA (NM_001017405) fusion: c.2327:FGFR3_c.457-2460:MAEAdup Protein Fusion: mid-exon {FGFR3:MAEA}	NA
EVT-P-001 0465-T01-I M5-42	P-0010 465-T0 1-IM5		FGFR 3-NSD2	NSD2	unknown	five_prime	1807917	13	653	True	disrupted	0.6570397111913358	True			Protein tyrosine and serine/threonine kinase	FGFR3 (NM_000142) - WHSC1 (NM_001042424) rearrangement:c.1959+17:FGFR3_c.-29-12740:WHS C1del Transcript fusion (FGFR3-WHSC1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 3954-T01-I M5-43	P-0013 954-T0 1-IM5		FGFR 3-NSD 2	NSD 2	unknown	threep _prime	180892 9	17	787	False	lost	0.0	False		Protein in tyro sine and serine/ threoni ne kinase		WHSC1 (NM_001042424) - FGFR3 (NM_000142) rearrangement: c.3255+4: WHSC1_c.2361:FGFR3dup Protein fusion: mid-exon (WHSC1-FGFR3)	NA
EVT-P-002 4991-T01-I M6-44	P-0024 991-T0 1-IM6		FGFR 3-RO BO1	RO BO1	unknown	fivep _prime	180896 3	17	799	False	retained	1.0	False	Protein tyrosine and serine/ threoni ne kinase			FGFR3 (NM_000142) - ROBO1 (NM_002941) fusion: t(3;4)(p12.3;p16.3)(chr3:g.78737098::chr4:g.1 808963) Protein Fusion: mid-exon {FGFR3:ROBO1}	NA
EVT-P-002 2387-T02-I M6-48	P-0022 387-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	fivep _prime	180870 7	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+46:FGFR3 _c.1942-224:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 4466-T02-I M6-49	P-0044 466-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	fivep _prime	180877 3	17	759	True	retained	1.0	False	Protein tyrosine and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-70:FGFR3_ c.1645-159:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 5884-T01-I M6-50	P-0055 884-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	fivep _prime	180888 6	17	773	False	retained	1.0	False	Protein tyrosine and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2318:FGFR3_c.1 942-928:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 0941-T02-I M6-51	P-0000 941-T0 2-IM6		FGFR 3-TAC C3	TAC C3	unknown	fivep _prime	180889 4	17	776	False	retained	1.0	False	Protein tyrosine and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2326:FGFR3_c.1 942-988:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-005 6846-T01-I M6-52	P-0056 846-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	fivep _prime	180869 8	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3 _c.1942-426:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 3694-T03-I M6-53	P-0053 694-T0 3-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180891 1	17	781	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1 942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-005 3694-T02-I M6-54	P-0053 694-T0 2-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180891 1	17	781	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1 942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-001 6989-T02-I M6-55	P-0016 989-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180871 8	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+57:FGFR3 _c.1645-139:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 3694-T01-I M6-57	P-0053 694-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180891 1	17	781	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2343:FGFR3_c.1 942-151:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-005 2352-T02-I M6-59	P-0052 352-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180876 4	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-79:FGFR3_ c.1837-58:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 2324-T01-I M6-60	P-0052 324-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180880 3	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-40:FGFR3_ c.1942-486:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 9195-T01-I M7-61	P-0059 195-T0 1-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180884 1	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-2:FGFR3_c .1645-51:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 0831-T01-I M7-62	P-0060 831-T0 1-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180882 7	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-16:FGFR3_ c.1941+414:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-006 0879-T01-I M7-63	P-0060 879-T0 1-IM7		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180894 5	17	793	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2377:FGFR3_c.1 942-241:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-001 1503-T01-I M5-64	P-0011 503-T0 1-IM5		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180896 7	17	800	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 10-16): c.2399:FGFR3_c.1836+75 :TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-005 1755-T01-I M6-65	P-0051 755-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180880 6	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-37:FGFR3_ c.1942-333:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 0534-T01-I M5-66	P-0010 534-T0 1-IM5		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180885 9	17	764	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 10-16): c.2363:FGFR3_ c.1784:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-001 0096-T03-I M6-67	P-0010 096-T0 3-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180892 5	17	786	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2357:FGFR3_c.1 941+378:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-006 0879-T02-I M7-68	P-0060 879-T0 2-IM7		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180894 5	17	793	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2377:FGFR3_c.1 942-241:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 2326-T01-I M7-69	P-0062 326-T0 1-IM7		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180891 3	17	782	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2345:FGFR3_c.1 386-12:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-005 1602-T01-I M6-70	P-0051 602-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180888 4	17	772	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2316:FGFR3_c.1 942-997:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-005 1575-T01-I M6-71	P-0051 575-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180872 4	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+63:FGFR3 _c.1461+64:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 1267-T01-I M6-72	P-0051 267-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180867 2	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+11:FGFR3 _c.1645-143:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 1173-T02-I M6-73	P-0051 173-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180871 6	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3 _c.1941+61:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-006 3306-T01-I M7-74	P-0063 306-T0 1-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180876 4	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-79:FGFR3_ c.1461+18:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-006 3929-T01-I M7-75	P-0063 929-T0 1-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180878 2	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-61:FGFR3_ c.1941+145:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 1724-T01-I M6-76	P-0051 724-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180876 0	17	759	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_c.1942-112:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 0757-T02-I M7-77	P-0050 757-T0 2-IM7		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180866 0	16	758	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2273:FGFR3_c.1572:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 6879-T01-I M5-78	P-0006 879-T0 1-IM5		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180896 7	17	800	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR3 exons 1 to 18 fused with TACC3 exons 11-16) : c.2399:FGFR3_c.1941+196:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-002 5859-T01-I M6-79	P-0025 859-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180099 7	2	42	False	lost	0.0	False		Protein tyrosine and serine/ threonine kinase		FGFR3 (NM_000142) - TACC3 (NM_006342) rearrangement: c.126:FGFR3_c.1941+33:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-005 0136-T01-I M6-80	P-0050 136-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180892 2	17	785	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2354:FGFR3_c.2040:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-002 7054-T03-I M6-81	P-0027 054-T0 3-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180889 8	17	777	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-002 7054-T05-I M6-82	P-0027 054-T0 5-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180889 8	17	777	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1942-902:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 7054-T07-I M7-83	P-0027 054-T0 7-IM7		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180889 8	17	777	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1 942-902:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 9741-T02-I M6-84	P-0049 741-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180883 0	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-13:FGFR3_ c.1942-479:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 9502-T01-I M6-85	P-0049 502-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180872 6	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+65:FGFR3_ c.1941+538:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 8985-T01-I M6-86	P-0048 985-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180882 7	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-16:FGFR3_ c.1788:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 8947-T01-I M6-87	P-0048 947-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180889 8	17	777	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2330:FGFR3_c.1 941+104:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 6660-T02-I M6-88	P-0006 660-T0 2-IM6		FGFR 3-TAC C3	TAC C3	out-of-frame	five_prime	180867 8	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+17:FGFR3_ c.1591+1556:TACC3dup Protein Fusion: out of frame {FGFR3:TACC3}	NA
EVT-P-004 7604-T01-I M6-89	P-0047 604-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180869 2	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3_ c.1941+484:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 5507-T01-I M6-90	P-0045 507-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808733	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3_c.1942-413:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 5382-T01-I M6-91	P-0045 382-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808903	17	779	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2335:FGFR3_c.1942-101:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 5753-T01-I M6-92	P-0047 573-T01-IM6		FGFR3-TACC3	TACC3	unknown	five_prime	1808844	17	759	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2276:FGFR3_c.1942-892:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-006 4650-T01-I M7-93	P-0064 650-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808781	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-62:FGFR3_c.1837-81:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-006 5466-T01-I M7-94	P-0065 466-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808795	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-48:FGFR3_c.1942-685:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-000 5361-T01-I M5-96	P-0005 361-T01-IM5		FGFR3-TACC3	TACC3	unknown	five_prime	1803212	4	188	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 5 fused to TACC3 exon 14) : c.564:FGFR3_c.2223+249:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-000 5128-T01-I M5-97	P-0005 128-T01-IM5		FGFR3-TACC3	TACC3	in-frame	five_prime	1808814	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3(NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon 17 fused to TACC3 exon 11) : c.2275-29:FGFR3_c.1942-314:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 2266-T01-I M6-98	P-0032 266-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180874 5	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+84:FGFR3 _c.1645-165:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 6238-T03-I M7-99	P-0016 238-T0 3-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180870 6	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+45:FGFR3 _c.1942-654:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 7604-T02-I M6-100	P-0047 604-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180869 2	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+31:FGFR3 _c.1941+484:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-006 6401-T01-I M7-101	P-0066 401-T0 1-IM7		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180893 0	17	788	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2362:FGFR3_c.1 942-935:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 4694-T01-I M5-102	P-0004 694-T0 1-IM5		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180888 1	17	771	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3(NM_000142) -TACC3(NM_006342) Fusion (FGFR3 exon 17 fused with TACC3 exon 9) : c.2313:FGFR3_c.1837-8 8:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-000 3853-T01-I M5-103	P-0003 853-T0 1-IM5		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180868 0	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-000 3827-T01-I M5-104	P-0003 827-T0 1-IM5		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180871 4	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.1941+232:TACC 3_c.2274+53:FGFR3dup Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 6580-T03-I M6-105	P-0046 580-T0 3-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180871 6	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3 _c.1943:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 6580-T02-I M6-106	P-0046 580-T0 2-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180871 6	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+55:FGFR3 _c.1943:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 5507-T04-I M7-107	P-0045 507-T0 4-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180873 3	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3 _c.1942-413:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 7126-T01-I M6-108	P-0037 126-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180887 8	17	770	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2310:FGFR3_c.1 837-16:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-003 8253-T01-I M6-109	P-0038 253-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180885 2	17	762	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) -TACC3 (NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon8) : c.2284:FGFR3_c.1644+1 06:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-003 8318-T01-I M6-110	P-0038 318-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180875 6	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-87:FGFR3_ c.1462-57:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 8414-T01-I M6-111	P-0038 414-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180886 9	17	767	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2301:FGFR3_c.1 941+974:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 9109-T01-I M6-112	P-0039 109-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180876 0	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_ c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 109-T02-I M6-113	P-0039 109-T0 2-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180876 0	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-83:FGFR3_ c.162+65:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 9116-T01-I M6-114	P-0039 116-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180869 8	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+37:FGFR3_ c.944:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-003 9138-T02-I M6-115	P-0039 138-T0 2-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180864 5	16	753	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2258:FGFR3_c.1 941+335:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-003 9166-T01-I M6-116	P-0039 166-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180888 3	17	772	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2315:FGFR3_c.1 942-61:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-006 7303-T01-I M7-117	P-0067 303-T0 1-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180881 2	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-31:FGFR3_ c.1941+863:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 9655-T01-I M6-118	P-0039 655-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180888 7	17	773	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2319:FGFR3_c.1 836+86:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003279-T02-I M6-119	P-0003279-T02-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) rearrangement: c.2274+47:FGFR3_c.1941+850:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004543-T01-I M6-120	P-0040543-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808696	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+35:FGFR3_c.1836+39:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0067315-T01-I M7-121	P-0067315-T01-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808877	17	770	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2309:FGFR3_c.1941+162:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0067602-T01-I M7-122	P-0067602-T01-IM7		FGFR3-TACC3	TACC3	in-frame	five_prime	1808775	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-68:FGFR3_c.1836+111:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0047905-T01-I M6-123	P-0047905-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808802	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-41:FGFR3_c.1837-96:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0042157-T06-I M7-124	P-0042157-T06-IM7		FGFR3-TACC3	TACC3	unknown	five_prime	1808855	17	763	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2287:FGFR3_c.1942-923:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0042393-T01-I M6-125	P-0042393-T01-IM6		FGFR3-TACC3	TACC3	in-frame	five_prime	1808709	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+48:FGFR3_c.1644+132:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 2515-T01-I M6-126	P-0042 515-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180884 1	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-2:FGFR3_c .1942-418:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-004 3076-T01-I M6-127	P-0043 076-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180892 6	17	786	False	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2358:FGFR3_c.2 019-34:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 4057-T01-I M6-128	P-0044 057-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180871 4	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+53:FGFR3 _c.1645-189:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 6624-T01-I M6-130	P-0016 624-T0 1-IM6		FGFR 3-TAC C3	TAC C3	unknown	five_prime	180873 3	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 17 fused to TACC3 exon 5): c .2274+72:FGFR3_c.1387: TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-004 5507-T03-I M7-131	P-0045 507-T0 3-IM7		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180873 3	16	758	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+72:FGFR3 _c.1942-413:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-005 4391-T01-I M6-132	P-0054 391-T0 1-IM6		FGFR 3-TAC C3	TAC C3	in-frame	five_prime	180878 9	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2275-54:FGFR3_ c.1941+447:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-000 6429-T01-I M5-133	P-0006 429-T0 1-IM5		FGFR 3-TNI P2	TNI P2	unknown	five_prime	180880 1	17	759	True	retained	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) -TNIP2 (NM_024309) rearrangement: c.2275-42: FGFR3_c.147:TNIP2inv Protein fusion: mid-exon (FGFR3-TNIP2)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006429-T02-I M6-134	P-0006429-T02-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808753	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-90:FGFR3_c.195:TNIP2inv Protein Fusion: mid-exon {FGFR3:TNIP2}	NA
EVT-P-0050971-T01-I M6-135	P-0050971-T01-IM6		FGFR3-TNIP2	TNIP2	unknown	five_prime	1808827	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TNIP2 (NM_024309) fusion: c.2275-16:FGFR3_c.252:TNIP2inv Protein Fusion: mid-exon {FGFR3:TNIP2}	NA
EVT-P-0012257-T01-I M5-136	P-0012257-T01-IM5		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808759	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16.3;q12.2)(chr4:g.1808759::chr16:g.55556795) Protein fusion: in frame {FGFR3-LPCAT2}	NA
EVT-P-0012257-T01-I M6-137	P-0012257-T03-IM6		LPCAT2-FGFR3	LPCAT2	in-frame	five_prime	1808708	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - LPCAT2 (NM_017839) rearrangement: t(4;16)(p16.3;q12.2)(chr4:g.1808708::chr16:g.55556796) Protein Fusion: in frame {FGFR3-LPCAT2}	NA
EVT-P-0024146-T01-I M6-139	P-0024146-T01-IM6		NSD2-FGFR3	NSD2	in-frame	three_prime	1801414	3	127	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			WHSC1 (NM_001042424) - FGFR3 (NM_000142) rearrangement: c.761-178:WHSC1_c.380-60:FGFR3 dup Protein Fusion: in frame {WHSC1:FGFR3}	NA
EVT-P-0017789-T01-I M6-141	P-0017789-T01-IM6		PTBP1-FGFR3	PTBP1	in-frame	five_prime	1808676	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - PTBP1 (NM_002819) rearrangement: t(4;19)(p16.3;p13.3)(chr4:g.1808676::chr19:g.802509) Protein Fusion: in frame {FGFR3:PTBP1}	NA
EVT-P-0033176-T01-I M6-142	P-0033176-T01-IM6		RBM20-FGFR3	RBM20	unknown	five_prime	1803168	4	174	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR3 (NM_000142) - RBM20 (NM_001134363) rearrangement: t(4;10)(p16.3;q25.2)(chr4:g.1803168::chr10:g.112544434) Protein Fusion: mid-exon {FGFR3:RBM20}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 6563-T01-I M6-143	P-0036 563-T0 1-IM6		TACC 3-FGF R3	TAC C3	out- of-fr ame	five _pr im e	180489 0	7	311	True	lost	0.0	False		Prote in tyr osine and s erine/ threo nine kinas e		FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion(FGFR3 exon7 fused with TACC3 exon4) : c.931-529:FGFR3_c.305 +1845:TACC3dup Protein Fusion: out of frame {FGFR3:TACC3}	NA
EVT-P-000 7131-T01-I M5-144	P-0007 131-T0 1-IM5		TACC 3-FGF R3	TAC C3	in-fr ame	five _pr im e	180876 4	17	759	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 with TACC3 exons 10-16) : c.2275-79:FGFR3_c.183 7-97:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-002 8647-T01-I M6-145	P-0028 647-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-fr ame	five _pr im e	180873 1	16	758	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2274+70:FGFR3 _c.1942-150:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 0871-T01-I M6-146	P-0030 871-T0 1-IM6		TACC 3-FGF R3	TAC C3	unk now n	five _pr im e	180885 6	17	763	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2288:FGFR3_c.1 608:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 0289-T01-I M3-147	P-0000 289-T0 1-IM3		TACC 3-FGF R3	TAC C3	unk now n	five _pr im e	180610 9	8	376	False	lost	0.0	False		Prote in tyr osine and s erine/ threo nine kinas e		FGFR3 (NM_000142) TACC3 (NM_006342) Duplication (c.1128_c.1592-1871dup) Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-003 2432-T01-I M6-148	P-0032 432-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-fr ame	five _pr im e	180867 4	16	758	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+13:FGFR3 _c.1837-134:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 8321-T01-I M6-149	P-0028 321-T0 1-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180788 4	13	648	False	disrupted	0.6389891 69675090 2	True			Protein tyrosin e and s erine/th reonine kinase	FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-14 fused to TACC3 exons 7-16): c.1943:FGFR3_c.1 591+377:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-002 7211-T01-I M6-151	P-0027 211-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180873 5	16	758	True	retained	1.0	False	Protein tyrosin e and s erine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 10-16): c.2274+74:FGFR3 _c.1836+111:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-002 7183-T02-I M6-152	P-0027 183-T0 2-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180867 3	16	758	True	retained	1.0	False	Protein tyrosin e and s erine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR1 exons 1-17 fused with TACC3 exons 11-16) : c.2274+12 :FGFR3_c.1942-636:TAC C3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 1155-T01-I M6-153	P-0031 155-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180871 5	16	758	True	retained	1.0	False	Protein tyrosin e and s erine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+54:FGF R3_c.1644+125:TACC3du p Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-002 6783-T01-I M6-154	P-0026 783-T0 1-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180884 8	17	760	False	retained	1.0	False	Protein tyrosin e and s erine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16) : c.2280:FGFR3_c.1942-7 8:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-002 4954-T02-I M6-155	P-0024 954-T0 2-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180875 5	17	759	True	retained	1.0	False	Protein tyrosin e and s erine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3 _c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_oint_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-002 4954-T01-I M6-156	P-0024 954-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-fr ame	five _pr im e	180875 5	17	759	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 11-16): c.2275-88:FGFR3 _c.1942-76:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-000 2351-T01-I M3-157	P-0002 351-T0 1-IM3		TACC 3-FGF R3	TAC C3	unk now n	five _pr im e	180887 1	17	768	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) and TACC3 (NM_006342) duplication (71843bp): FG FR3:c.2303_TACC3:c.162 0dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-002 4346-T01-I M6-158	P-0024 346-T0 1-IM6		TACC 3-FGF R3	TAC C3	unk now n	five _pr im e	180885 9	17	764	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 6-16): c.2291:FGFR3_c.1 461+69:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-003 3514-T01-I M6-159	P-0033 514-T0 1-IM6		TACC 3-FGF R3	TAC C3	unk now n	five _pr im e	180888 3	17	772	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 4-16): c.2315:FGFR3_c.9 27:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-002 3261-T01-I M6-160	P-0023 261-T0 1-IM6		TACC 3-FGF R3	TAC C3	unk now n	five _pr im e	180892 7	17	787	False	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to exons 11-16 of TACC3): c.2359:FGFR3_c .1941+673:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-002 2569-T01-I M6-161	P-0022 569-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-fr ame	five _pr im e	180870 5	16	758	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused with TACC3 exons 8-16. This includes the kinase domain of FGFR3.): c.2274+44:FGF R3_c.1645-194:TACC3du p Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 0355-T01-I M6-162	P-0020 355-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180880 7	17	759	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 11-16): c.2275-36:FGFR3 _c.1942-312:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 9730-T01-I M6-163	P-0019 730-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180875 8	17	759	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16):c.22 75-85:FGFR3_c.1836+25: TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-003 4183-T01-I M6-164	P-0034 183-T0 1-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180891 4	17	782	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 7-16): c.2346:FGFR3_c.1 591+1975dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-001 7551-T01-I M5-165	P-0017 551-T0 1-IM5		TACC 3-FGF R3	TAC C3	unknown	five_prime	180887 7	17	770	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused with TACC3 exons 10-16): c.2309:FGFR3_c. 1836+8:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-001 7444-T01-I M6-166	P-0017 444-T0 1-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180886 4	17	766	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused with TACC3 exons 11-16): c.2296:FGFR3_c. 1941+70:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-001 7056-T01-I M6-167	P-0017 056-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180867 1	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 11-16): c.22 74+10:FGFR3_c.1941+88 1:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 6989-T01-I M6-168	P-0016 989-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	1808718	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused to TACC3 exons 8-16): c.2274+57:FGFR3_c.1645-139:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-000 1453-T01-I M3-169	P-0001 453-T0 1-IM3		TACC 3-FGF R3	TAC C3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-000 1453-T01-I M3-170	P-0001 453-T0 1-IM3		TACC 3-FGF R3	TAC C3	unknown	five_prime	1808946	17	793	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-001 6238-T01-I M6-171	P-0016 238-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	1808706	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 17 fused to TACC3 exon 11): c.2274+45:FGFR3_c.1942-654:TACC3inv Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-001 5863-T01-I M6-173	P-0015 863-T0 1-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	1808880	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2312:FGFR3_c.1644+147:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-001 5195-T01-I M6-174	P-0015 195-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	1808757	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon17 fused to TACC3 exon8): c.2275-86:FGFR3_c.1645-28:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 5024-T01-I M6-175	P-0015 024-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180821 0	15	677	True	disrupted	0.7436823 10469314 1	True			Protein tyrosine and serine/ threonine kinase	FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exon 16 fused to TACC3 exon 5) : c.2031-63:FGFR3_c.1385 +14:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-001 4530-T01-I M6-176	P-0014 530-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180871 5	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) Fusion (FGFR3 exons 1-17 fused with TACC3 exons 6-16) : c.2274+54:F GFR3_c.1462-52:TACC3d up Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-001 4237-T01-I M6-177	P-0014 237-T0 1-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180895 8	17	797	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 10-16): c.2390:FGFR3_c.1836+80 :TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-001 2654-T01-I M5-178	P-0012 654-T0 1-IM5		TACC 3-FGF R3	TAC C3	unknown	five_prime	180893 5	17	789	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) -TACC3 (NM_006342)fusion: c.236 7:FGFR3_c.1645-121:TA CC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-001 2373-T01-I M5-179	P-0012 373-T0 1-IM5		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180873 0	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 with TACC3 exons 10-16): c.2274+69:FGFR3_c.18 37-70:TACC3dup Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-003 4666-T01-I M6-180	P-0034 666-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180872 1	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+60:FGFR3 _c.1941+226:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

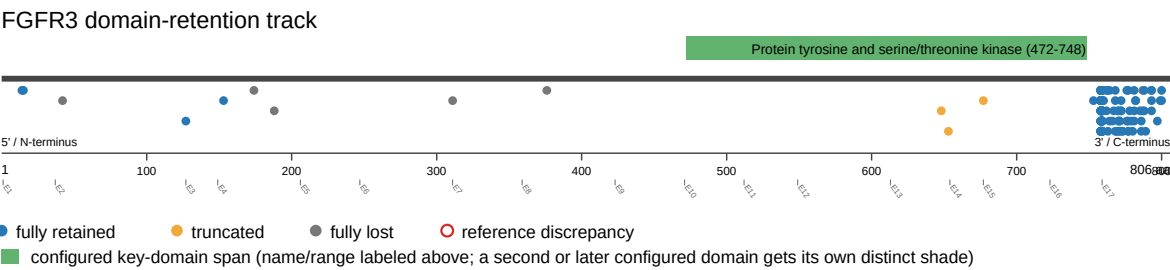
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 6305-T01-I M6-181	P-0036 305-T0 1-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180890 2	17	778	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 - (NM_000142) TACC3 (NM_006342) Fusion (FGFR3 exon18 fused with TACC3 exon3) : c.2334:FGFR3_c.274:TA CC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 8649-T01-I M5-182	P-0008 649-T0 1-IM5		TACC 3-FGF R3	TAC C3	unknown	five_prime	180887 8	17	770	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 7-16 of TACC3): c.2310:F GFR3_c.1645-1546:TACC 3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-002 9600-T01-I M6-183	P-0029 600-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-frame	five_prime	180869 7	16	758	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+36:FGFR3 _c.1644+158:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-000 7082-T03-I M6-184	P-0007 082-T0 3-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180887 2	17	768	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3(NM_000142) - TACC3(NM_006342) Fusion (FGFR3 exon 18 fused to TACC3 exon 8) : c.2304:FGFR3_c.1644+11 8:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 7082-T02-I M6-185	P-0007 082-T0 2-IM6		TACC 3-FGF R3	TAC C3	unknown	five_prime	180887 2	17	768	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-18 of FGFR3 fused to exons 8-16 of TACC3): c.2304:F GFR3_1644+118:TACC3d up Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-000 5966-T01-I M5-186	P-0005 966-T0 1-IM5		TACC 3-FGF R3	TAC C3	unknown	five_prime	180885 3	17	762	False	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1 to 17 with TACC3 exons 11 to 16) : c.2285:FGFR3_c.1 941+423:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005862-T01-I M5-187	P-0005862-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808707	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (exons 1-17 of FGFR3 fused to exons 14-16 of TACC3): c.2274+46:FGFR3_c.2223+896:TACC3dup. Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0004902-T01-I M5-189	P-0004902-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808906	17	780	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 with TACC3 exons 11-16): c.2338:FGFR3_c.1941+300TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0003677-T01-I M5-190	P-0003677-T01-IM5		TACC3-FGFR3	TACC3	unknown	five_prime	1808931	17	788	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			c.2363:FGFR3_c.1784:TACC3dup Protein fusion: mid-exon (FGFR3-TACC3)	NA
EVT-P-0003097-T01-I M5-191	P-0003097-T01-IM5		TACC3-FGFR3	TACC3	in-frame	five_prime	1808841	17	759	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (FGFR3-TACC3)	NA
EVT-P-0036396-T01-I M6-192	P-0036396-T01-IM6		TACC3-FGFR3	TACC3	in-frame	five_prime	1808747	16	758	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion: c.2274+86:FGFR3_c.162+46:TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA
EVT-P-0016735-T01-I M6-193	P-0016735-T01-IM6		TACC3-FGFR3	TACC3	unknown	five_prime	1808879	17	771	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-18 fused to TACC3 exons 8-16): c.2311:FGFR3_c.1644+71:TACC3dup Protein Fusion: mid-exon {FGFR3:TACC3}	NA
EVT-P-0002862-T01-I M3-194	P-0002862-T01-IM3		TACC3-FGFR3	TACC3	unknown	five_prime	1808873	17	769	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: mid-exon (FGFR3-TACC3)	NA

event_id	sampl e_id	pa tie nt _i d	fusion _nam e	part ner _ge ne	fra me _sta tus	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-002 4463-T01-I M6-195	P-0024 463-T0 1-IM6		TACC 3-FGF R3	TAC C3	in-fr ame	five _pr ime	180880 1	17	759	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			FGFR3 (NM_000142) - TACC3 (NM_006342) fusion (FGFR3 exons 1-17 fused in-frame with TACC3 exons 10-16):c.22 75-42:FGFR3_c.1836+11: TACC3dup Protein Fusion: in frame {FGFR3:TACC3}	NA

Figures

domain retention outliers



fusion schematic

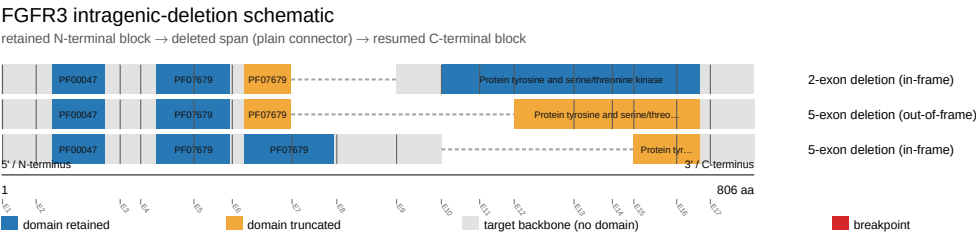
FGFR3 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FGFR3 portion is domain-colored, exon ticks)



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 52 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

